

Title: ICA Artifact Removal

Category: Preprocessing

Libraries: MNE, Pandas

1. Objective: To separate and automatically remove physiological artifacts (such as eye blinks, muscle movements, or heartbeats) from the brain wave data without deleting the underlying healthy EEG signal. Unlike Epoch Rejection which throws away the entire time segment, ICA acts like a surgical filter, extracting only the noise and leaving the clean brain activity intact for the AI model to learn from.

2. How it Works (Mathematical Concept & Algorithm) The algorithm uses MNE's FastICA implementation and operates in four distinct stages:

- **Step 0: Prerequisites (High-Pass Filter)** Before ICA can work, the data must be high-pass filtered (e.g., at 1.0 Hz) to remove "slow drifts" caused by sweat or head movements. Without this, the algorithm wastes components trying to model the drift instead of the blinks.
- **Step 1: Spatial Decomposition** FastICA transforms the mixed EEG signals from the scalp channels into a set of "Independent Components" (source signals). Mathematically, it assumes that brain activity and eye blinks are generated by physically distinct, statistically independent sources.
- **Step 2: Automatic Artifact Detection (The "Smart Proxy" Test)** The algorithm needs to identify which of these independent components represent eye blinks.
 - It looks for a frontal channel closest to the eyes (e.g., Fp1). If Fp1 is missing, a Smart Fallback mechanism automatically hunts for the next best channel (e.g., Fp2, Fz, or channel_1).
 - It calculates a Pearson correlation (Z-score) between every ICA component and the chosen frontal channel.
 - If a component's correlation exceeds the Threshold Parameter (e.g., Z-score > 1.5), it is flagged as a blink artifact.
- **Step 3: Signal Reconstruction** The algorithm "zeros out" (drops) the flagged bad components and multiplies the remaining good components back into the original channel space, resulting in a clean signal.

3. Example (Component Evaluation Scenario):

Scenario: We decomposed a signal into 4 components. Our Threshold is set to 1.5, and our EOG proxy channel is Fz.

Component ID	Correlation (Z-Score) with Fz	Status	Action	Reason
Component 0	4.2	Fail	Zeroed Out	Exceeds threshold ($4.2 > 1.5$). Strong eye blink signature.
Component 1	0.4	Pass	Keep	Normal brainwave activity (low correlation with eyes).
Component 2	1.8	Fail	Zeroed Out	Exceeds threshold ($1.8 > 1.5$). Weak or partial blink.
Component 3	0.9	Pass	Keep	Normal brainwave activity.

Result: Components 0 and 2 are deleted. The EEG data is reconstructed using only Components 1 and 3.